AI Project: Automated Recognition of Unknown Marine Species

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1 Introduction

Due to current advancements in autonomous and artificially intelligent technologies, tasks humankind once found unfeasible, are now within grasp. This technological progress one can currently find, motivated this search for a project that would require great manpower, time, and nonstop, systematic work that could, instead, be automated. There exists a plethora of scientific disciplines and tasks, which could benefit from the addition of an artificial intelligent component and system. In the two areas this report will cover, automation is overdue and greatly needed.

Exploring our oceans is a task not even modern technology has completely dominated and is a task humans have strived to undertake since the beginning of sea travel. Even though oceans cover over 70 percent of the Earth's surface, scientists expect to have explored less than 20 percent of them. [18] Automating ocean exploration and analysis would be quicker, more cost effective and overall a more efficient approach. Specifically, humans could stand to benefit from the cataloging of marine species. Many new species identified today are found in the ocean and there is an overabundance of marine environments left to be thoroughly explored. Analysis of marine species holds a wealth of intrinsic value and could lead to advances in a variety of fields such as medicine, biochemistry, and evolutionary biology. [UA18]

This project will focus on the recognition and identification of marine species, which is a subset of the field alpha taxonomy. "Alpha Taxonomy is the discipline of detecting, describing, and classifying new species, as well as revising the classification of previously described species."[Dic10] The arduous manual task of species identification, confirmation, and classification can only be officially done by few experts and is known to be inconsistent. [NC10] Alpha taxonomy is a neglected area of study that is in high demand in various scientific fields such as biology, ecology and conservation. [NC10] Furthermore, better accessibility of identification is needed, so it can be used across diverse scientific disciplines. However, it remains drastically underfunded and understudied.

After the realization that these two fields could drastically benefit from the automation of various tasks, this project decided to focus on the recognition of undiscovered marine species. This report proposes an answer to the question, what an automated system architecture for the recognition of previously unknown marine species could look like. It describes a novel hybrid intelligence system that utilizes a machine learning algorithm to recognize previously undiscovered or unknown marine species. The proposed algorithm is best suited for use on a ship or underwater vessel for real-time species identification in the ocean using a camera apparatus. This paper aims to analyze the potential viability and plausibility of our algorithm.

To the best of the authors' knowledge, automated recognition of previously unknown species has not been done before. The combination of machine learning algorithms and autonomous identification of known species is just being explored.[NC10] A variety of projects aim to utilize image identification to classify species using algorithms, but they tend to focus on a single species or class of animal and range in accuracy and viability. This report aims to contribute the concept of using an artificially intelligent algorithm to detect novelties, i.e. objects it has not previously been trained to identify, specifically in the field of alpha taxonomy. The applications for such a system are numerous and, although, this project is centered around an ocean environment, it could potentially be adapted for other settings.

2 Problem and Domain

Alpha taxonomy requires immediate assistance. Currently, it is an underfunded field dependent on the expertise of few and in demand by many [NC10]. It necessitates a transformation into an automated program, that is cost and time efficient and widely accessible. Taxonomists that practice alpha taxonomy are the basis to a plethora of scientific disciplines such as applied biology, ecology, agriculture and conservation [NC10]. Species identification demands in various fields of sciences are ever growing with the introduction of easier methods of collecting data, but still the alpha taxonomy field keeps shrinking. Due to financial constraints, there are few experts officially certified to classify new species and identify those known, which does not meet the demand in various fields [NC10]. There are approximately four to six thousand professional taxonomists worldwide, only a subset of whom practice alpha taxonomy [NC10].

This field requires the establishment of a stable foundation on which other scientific disciplines can build. Presently, there is no precedent of independent testing and verification of the accuracy of identification that taxonomists produce, due in part to the underfunding and lack of support the field currently has [NC10]. There are few studies comparing the identification of species by various experts in the field of alpha taxonomy but one of the few conducted produced the concerning result that accuracy ranges in identification and classification of species with an average of only 72 percent consistency between experts [NC10]. Disputes between experts, however, are unsurprising due to the inconsistencies existing in this field and its reliance on human visual cognition. The lack of verification, inability to establish reliable references, and the human error that runs rampant makes alpha taxonomy a prime candidate for automation.

Methods for automation of alpha taxonomy have been attempted before with little success. DNA barcode identification was proposed as a potential way to identify species using a genetic mapping-based approach and a database of all known species DNA makeup. However, this method was found to be a major cost and inaccessible to scientists without genetic knowhow [NC10]. Alpha taxonomy normally relies on pattern and shape recognition, or morphological diagnoses, when performed manually, but that could be modified to be done using image recognition machine learning algorithms [Ger18].

As previously stated, humankind has a magnitude of work left to do in order to explore Earth's oceans. Marine environments retain a wealth of potential information that remains untapped. It is presently estimated that 91 percent of ocean species are left to be discovered [UA18]. For instance, species have been recently found belonging to a new domain of life, the Archaea, an ancient lifeform most closely related to the first life on Earth [WWF20]. The specialized adaptations of these creatures could expand our understanding of evolution, speciation and biochemistry, which in turn could enable new medical advances [Ger18].

Conquering Earth's oceans has always been established as a scientific goal but yet so much work is left to be done. This field requires journeys into treacherous waters, systematic work, and an abundance of manual hours. Automation of the exploration of our oceans is the next logical step. Therefore, the main objective of this project is to discover and classify new marine species by using state-of-the-art technologies like machine learning algorithms. Henceforth, it aims to fill this gap in human knowledge and conquer the anonymity of marine environments and species.

3 Literature and Knowledge Acquisition

Although novelty detection machine learning algorithms have not yet been used in the field of alpha taxonomy, image based species identification does exist for known species. Furthermore, novelty detection has already been used in object recognition. We hope to combine preexisting concepts into a new and revolutionary system for recognizing previously uncatalogued marine species. Thus, we investigated and established what currently exists in both fields, in order to assist in the installment of a solid foundation for our system. We perused google scholar with various queries such as automated species identification, species image recognition machine learning, and image-based novelty detection machine learning algorithms. These allowed us to find a plethora of research papers and articles that informed us of what has been done, what needs to be done, and how we could potentially set up our system. Additionally, we consulted and conducted interviews with two local experts; one a computer scientist with a specialization in algorithms and the other the head of technology at an oceanographic engineering firm. Below we will provide a quick summary of the essence of the papers, articles and interviews we consulted and what they say has been previously done.

3.1 Machine learning for image based species identification by Wäldchen, Jana, and Patrick Mäder

This paper begins by emphasizing the importance of taxonomic research in various fields of science. It says due to the overabundance of image data available that image-based machine learning algorithms for identification should be explored. The authors then describe how computer vision works and its relevance to this task. Additionally, they illustrate the necessity of deep learning neural networks and how they could replace the need for manual characteristic derivation with automation. It, then, provides an overview of the beginning of the establishment machine learning in image-based species identification. They site a plethora of work and how most are done. There exists two types of categories for automated species identification: lab-based investigations and field-based investigations. It then describes the multiple studies conducted using deep learning algorithms such as ResNet-10 and AlexNet which were used ti identify mammals, insects or fishes in various studies with a fairly decent accuracy, usually above 90 percent. They also describe the current app- based project conducted by Visipedia, the Cornell Laboratory of Ornithology, Cornell Tech and Caltech, called Merlin Photo ID. this project aims to to identify 650 of North America's most common bird species based on images using Google's TensorFlow deep learning platform and data from the eBird platform (available at `http://merlin.allaboutbirds.org/photo-id/`, successfully accessed October 21, 2020). This paper, also, listed other current projects using image-based species identification in the field like the iNaturalist.org project [Ger18].

3.2 Time to automate identification by MacLeod, Norman, Mark Benfield, and Phil Culverhouse

This article emphasizes the necessity of automating species identification and describes a couple projects that have done just this. The authors demonstrate that alpha taxonomy is the manual identification of species and it is only able to be done by a specific group of experts. It is a dying field that is also lacking funding, verification and consistency across experts. Identification of species is vital in multiple fields. Manual species identification also lacks accuracy and is the subject of much discussion between experts. Alternative methods like DNA bar coding are not easily scalable and expensive. New projects like DAISY and DiCANN are already providing as accurate identifications as experts on average [NC10].

3.3 Vehicles for deep sea exploration by John H. Steele, Steve A. Thorpe and Karl K. Turekian

We also investigated vehicles that could be used to best propagate our AI system in underwater environments. The deep sea, for instance, is a potential environment where experts predict previously undiscovered species could dwell. In the recent years, 2 types of remotely operated vehicles (ROV) and autonomous underwater vehicles (AUV) have been developed. With these deep submerge vehicles we can potentially expand our capabilities to measure, map and sample in secluded and inhospitable parts of the ocean.A hybrid remotely operated vehicle (HROV) is able to switch back and forth to operate as an ROV or an AUV on the same cruise. With this we can explore the deepest parts of the world's oceans, and we can even explore ice-covered oceans, such as the Arctic. AUV's are vehicles that are unoccupied and untethered. They are programmed to drift, glide or drive through the deep sea without real-time intervention from human operators. The data that is collected by the AUV are being recorded and transmitted via satellite when the AUV comes back to the surface of the sea. The advantage of AUVs compared to HOVs and ROVs is that these vehicles are more portable, and they can be deployed off many ships. AUV's are ideal for exploring large areas of the ocean, because it would take years to cover this by using ROVs or HOVs [Hum09].

3.4 Review of Novelty Detection Methods by Dubravko Miljković

We also read multiple very complex papers describing the very new concept of novelty detection in machine learning algorithms. This paper we found summarizes different methods and approaches of novelty. detection and mentions some possible applications: [Mil10]

- Identification of new or unknown data à abnormal pattern that the system is not aware of during training.
- Train the neural network on the normal training data and then detect novelties by analyzing input data.

This paper is a short review of Novelty Detection and its methods. According to this paper, Novelty (anomaly, outlier, exception) is a pattern in the data that does not conform to the expected behavior. Therefore, Novelty Detection refers to identifying novel or abnormal patterns embedded in a large amount of standard data. It gives a general design of a framework for such detection. This design combines various knowledge disciplines. Finally, it presents the application domains. Some of the patterns introduced: Point pattern, Collective pattern, Contextual pattern. Some of the methods introduced: Classification Based, Nearest Neighbor Based, Clustering Based, and Statistical approach. Some of the applications cited: Intelligent monitoring of high integrity systems, medical diagnostics, and video surveillance.

3.5 An approach to novelty detection applied to the classification of image regions by Sameer Singh and Markos Markou

The paper presents a new framework for novelty detection. The framework evaluates neural networks as adaptive classifiers that are capable of novelty detection and retraining on the basis of newly discovered information. Then apply our newly developed model to the application area of object recognition in video.

We detail the tools and methods needed for novelty detection such that data from unknown classes can be reliably rejected without any a prior knowledge of its characteristics. The rejected data is postprocessed to determine which samples can be manually labeled of a new type and used for retraining. We compare the proposed framework with other novelty detection methods and discuss the results of adaptive retraining of neural network to recognize further unseen data containing the newly added objects. [SM04]

3.6 Novelty detection in image recognition using IRF Neural Networks properties by Philippe Smagghe, Jean-Luc Buessler, Jean-Philippe Urban

Image Receptive Fields Neural Network (IRF-NN) is a variant of feedforward multi-layer perceptrons adapted to image recognition. It shows very fast training as well as robust and accurate results on supervised classification tasks. This paper presents another property of IRF-NN: responses of trained networks can be analysed to detect unknown images. IRF-NN also allows the use of the statistical properties of the neural output to detect novelty, and this paper shows it has successfully tested several criteria on large scale applications with tens of thousands of images. [SBU13]

3.7 Fast and Accurate Fish Detection Design with Improved YOLO-v3 Model and Transfer Learning by Kazim Raza and Hong Song

To identify new marine species, a faster real-time object algorithm is of prime importance. Besides, new species identification can prove to be quite challenging where labelled training data is scarce. Therefore, the need for a transfer learning approach for merging plentiful source-domain data with limited samples from possible new species to create a classifier that ideally performs nearly as well as if a lot of data were present. [RH20]

3.8 Detecting, Tracking and Counting Fish in Low Quality Unconstrained Underwater Videos by Concetto Spampinato, Yun-Heh Chen-Burger , Gayathri Nadarajan, Robert B. Fisher.

This paper presents an alternative approach by using an automated Video Processing (VP) system that analyses videos to identify interesting features. These videos are taken automatically and continuously by underwater video-surveillance cameras. It explains some of the difficulties faced when using cameras to obtain input for Machine Learning and proposes some techniques to process the images obtained to improve results. [Spa+08]

3.9 Interview with Marcel Walter

In order to further our understanding of this topic we conducted a couple interviews with local experts we found that were graciously willing to give us some advice. In this interview, we talked to Marcel Walter, a researcher in Computer Architecture at the University of Bremen in Germany. Currently, he is a final-year Ph.D. student with a specialization in algorithms for the design of emerging technologies. Marcel clarifies some of the state-of-the-art techniques in Machine Learning that could be useful in our proposed solution and the potential difficulties that we could encounter along the way. Moreover, from a technical perspective, he assesses how achievable our solution is. The full transcript of this interview will be found in our appendix for consultation.

3.10 Interview with Juan Gardner

In the next interview, we talked to Juan Gardner, who is the technical manager for Fugro Marine Germany GmbH. His company is specifically associated with doing hydrographic work for marine charting as well as full geophysical and geotechnical survey for subsea cables and power. Artificial Intelligence is most used in his field for editing and for target classification. He clarifies some of the vehicles that could host our AI. The most stereotypical is the larger science and research vessels holding the larger sonars. Larger vessels use lower frequency and lower resolution sonars but have the largest foundation of learning materials. Whereas the smaller AUVs use higher frequency sonars, which are more precise but have less available learning material. He mentions that AUVs are available globally, but the very large and more sophisticated units have less than a thousand globally. However nearly every single AUV has capacities that could be used in this application. He finally mentioned that there are already many low carbon footprint platforms collecting various data autonomously in the ocean, but it's most efficient to have larger vessel deploy multiple small AUVs as a force multiplier.

4 Solution

The main objective of this project is to discover new marine species by using state-of-the-art technologies such as machine learning algorithms. The study of species identification through machine learning is just being explored with a variety of convolutional neural network algorithms [NC10]. However, these systems are only able to detect a small number of preselected species and are not able to distinguish between discovered and undiscovered. Therefore, we propose to combine already existing methods and algorithms with novelty detection to identify unknown species in a hybrid intelligent system.

The general idea behind our system is to optimize on-site identification and detect new species by the process of elimination. Encountered marine life will be compared to our system's knowledge of known marine life forms. Once the AI encounters an animal, it will analyze its features and compare these to feature patterns of known species. A percentage is then being calculated, in which the animal differs from those attained from the database. If this percentage reaches a certain threshold, this species will be flagged as potentially undiscovered and a variety of data, such as pictures and videos, would report to scientists for further processing, making it a hybrid system. The algorithm alone cannot be certain if it has found a new species and, therefore, depends on human assistance. For instance, an abnormal pattern of features could also turn out to be a strange mutation.

The required database containing all marine creatures could for example be obtained from the *Oceanic Biodiversity Information System* (OBIS) [GS99]. Images of cataloged marine species from the OBIS could then train a convolutional neural network and thereby establish our machine learning system.

To ensure the accuracy of our system we must reduce the number of classes, our AI can identify. The expectation to identify all marine life is too exorbitant and would overwhelm our algorithm. Other projects in that area of expertise have limited their systems to a certain kind of animal [NC10]. However, that will not work for the purpose of new species identification. Therefore, we propose a location-based approach instead of a species-based approach, which builds upon already known locations of species and their migration patterns. If one is attempting to find new species in a certain area of the ocean, we will specifically focus on the input of images of the fauna previously found there, to make outliers more obvious.

Our machine learning algorithm is built to be employed in field settings, in this case, the ocean. After the initial establishment of our system and its training, our AI can be released into or set up in the environment, so it can hunt for undiscovered species. Input during this phase would come from the environment for on-site, real-time identification. This could be propagated by either a set of stationary cameras or a set of cameras attached to a drone-like apparatus. This apparatus could be either remote-controlled or even an autonomous underwater vehicle.

Once our system recognizes an unknown species, the underwater vehicle will follow it briefly to gather multiple images and notify scientists for confirmation and consultation. Using autonomous underwater vehicles (AUV), which do not rely on constant wireless communication with the surface, would be especially suited for deep-sea missions, where a plethora of new species could potentially be found. Vehicles with this kind of equipment and adaptations do already exist. There has even been a recent proposal of an improved, more efficient AUV from the SEA-Kit project, whose autonomous navigation system would be compatible with our AI. [SEA]

Henceforth, we propose building an algorithm, which is based on a convolutional neural network, which has been trained to recognize objects (focus on marine environment).

To identify objects in images and detect novelties representing unknown objects, our algorithm must obtain both a training phase (completing a stage of supervised learning) and a testing phase. Both will be elaborated in the following sections.

4.1 Training phase

At first, our algorithm must be trained on ground truth data for known objects. This data would be obtained by the already mentioned OBIS and can be further limited and specified based on the desired environmental location. These training data images will then get segmented and split into regions. The segmentation process prepares the images for the feature extraction. During this extraction, our machine learning algorithm could potentially store pixel values such as color and texture features for each segmented region.

4.2 Testing phase

The testing phase is similar to the first half of the training phase and represents the actual work the algorithm will perform regarding novelty detection. The testing phase can also be denoted as the environmental or field phase. In this phase, our AI system will be confronted with real time images segmented form live video streams. It will then apply its machine learning to recognize novelties, i. e., undiscovered species, in the environment it was placed. Data is attained through cameras, either on AUVs or stationary.

Image interpretation in underwater environments has a variety of obstacles caused by light dilution and water refraction. Therefore, our AI would also incorporate an image enhancement technique established through deep learning that has recently been developed and tested in order to emulate training phase data as much as possible [Iqb+07].

Thus, our system cross-references detected features of the encountered animals to known feature patterns of the classification network. If the object turns out to be an animal, the rejection filter will start working on searching for abnormal feature patterns. In other words: it will check if the given image of the marine animal has already been discovered at the current location, at all or is unclassified. All these decisions will be made by using the process of elimination and calculating percentages of overlapping features between classes and the given object. If the threshold defining an encountered animal's consistency with known species is breached, our algorithm will conclude it has detected an anomaly and notify scientists. Detected known animals will be classified by the neural network and add to the classification network by performing a cluster comparison with known distributions to isolate new classes. Retraining our algorithm constantly and on-site prevents the detection of an unknown species multiple times.

In conclusion, to implement the main objective of this project (identify unknown marine species) we are using the process of elimination and narrowing down the application with a location-based approach. In order to distinguish between undiscovered and discovered species, we calculate the percentage of overlapping feature patterns, in order to then conclude whether a species is anomalous to those established in the training phase as known. The machine

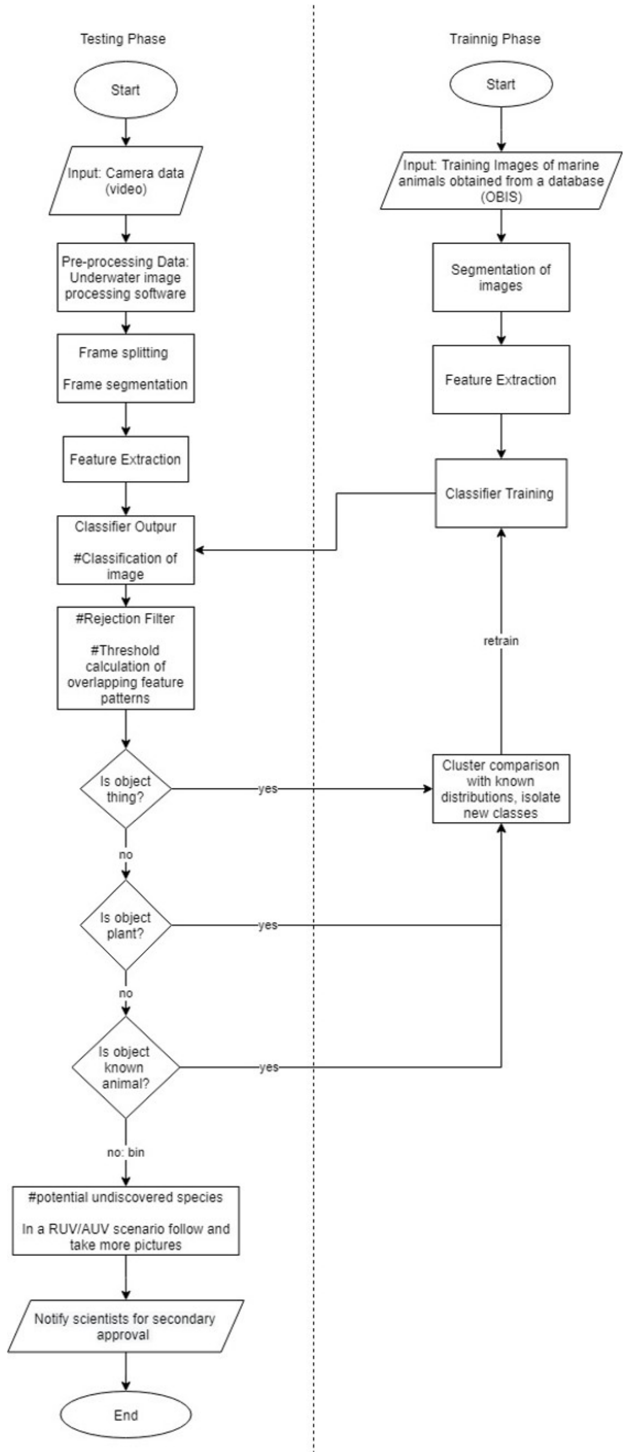

Fig. 1. Flowchart for our proposed algorithm illustrating the two phases: training and testing

learning system will obtain the required labeled data from the OBIS and unlabeled testing data from cameras attached to underwater vehicles.

5 Evaluation

5.1 Potential Experimental Validation

Initially, we engage in the training phase. In this experimental setting instead of training our system with all available species for our experimental locale, we withhold a subset of our labeled training data. Henceforth, though they would be previously cataloged species, they would remain novelties to our system. This method of experimentation is commonly known as cross validation. A potential experiment we could conduct could look like the following: The training data is obtained as a sequence of still images from the OBIS (Ocean Biodiversity Information System). A total of 1050 still images containing known marine species are chosen. For this experiment, the training data set consists of four classes: anemone-fish, star-fish, jellyfish, and sharks. During our training phase these images are inputted into our machine learning algorithm in order for it to establish an array of neural network layers that will analyze the morphological features in the species in the images.

Next, we will create a mock field setting in a laboratory aquarium. The testing data or field data is obtained as a sequence of video images captured by cameras placed underwater placed strategically in our aquarium. Our aquarium currently holds specific aquatic species: anemone-fish, star-fish, jellyfish, sharks, seahorse and salmon. Two of the marine species present are unknown to our model: seahorse and salmon. Video data in the aquarium will then be gathered and images would be sampled and analyzed.

Our experiment would have two field or test phases. In our first test phase we use a video sequence labeled V1 which contains footage of all the species in our tank. We would then manually label each image with certain species in them and create at table of the number of times each animal occurs in these images. This will then be used for comparison during our experiment. We will, then, compare the number of identifications our algorithm would make to our manual labels to create an accuracy vector. In the second phase, we generate a new training data set and a new video sequence and images. Before the second phase begins we would retrain our algorithm to include the species we withheld. The retraining is performed incrementally, and the retrained network is tested on video sequence V2 that contains the newly added species. The evaluation of this phase is based on the correct classification of objects in V2, using a similar technique to phase one. This would then be evaluated similarly for accuracy after the employment of our algorithm. The theoretical composition of the training data and test data sets are shown in Fig. 2.

The main aim of this analysis is to investigate the quality of the rejection filter, the accuracy of our model when identifying known species, detecting novelties, and isolating new classes.

	1st phase		2nd phase	
	Training	Testing	Training	Testing
Data	OBIS	V1	OBIS + V1	V2
Anomone-fish	450	150	450	100
Jellysfish	250	80	250	90
Star-fish	200	50	200	50
Shark	150	30	150	30
Seahorse	-	70	70	35
Salmon	-	50	50	15

Fig. 2. Data composition (i. e. number of images) in the training and testing phase. The Training phase images are taken from OBIS, while testing or field images are segments of the video taken by the camera apparatus in the aquarium.

5.2 Results

In order to validate our algorithmic solution, we expect the following experimental results:

1. High accuracy clustering using video sequence V1,
2. identification of novel species
3. High accuracy classification using video sequence V2 after retraining the neural network.

6 Conclusion

Advances in the discipline of Alpha Taxonomy are in high demand in various scientific fields and should therefore, not be neglected any longer. For instance, the discovery of new species influences biology, medicine and evolutionary science and the best is still to come. Thanks to the advancements in AI, we can finally make progress in conquering the ocean. A problem that has been bothering humans for centuries.

To advance the exploration of the oceans, our proposed machine learning algorithm will assist scientists in searching for undiscovered species, thus utilizing a hybrid intelligence system. This project implements already existing technologies in order to establish a new and revolutionary program . For instance, we can take advantage of already developed image classification systems and partner up with other research groups to access working AUVs.

Combining both novelty detection and machine learning algorithms, which can identify and classify objects in images, holds great potential. Our system will obtain its images of known species (labeled training data) from the OBIS. Images of objects to analyze will be received from an environmental input. This could be realized by a set of cameras attached to an underwater vehicle.

To increase efficiency, we will narrow down the wide range of known species from the OBIS with a location-based approach. Furthermore, we are using the process of elimination to classify objects in order to implement the main objective of this project: identifying new species. If a potential new species has been detected, images will get reported to scientist for further processing.

To validate our solution, we proposed a potential experiment consisting of training/retraining and testing phases. In the testing phase we are using unlabeled data to check the system's ability to distinguish between discovered and undiscovered species. The results expected were a high accuracy in the clustering mechanism, identification of novel species, and high precision of classification after retraining the neural network. If the algorithm does not match our expectations and creates errors, the machine learning system must be adapted. In conclusion, this solution proposes a more optimal way to recognize unknown marine species by using state-of-the-art technology such as Artificial Intelligence and novelty detection. Although it is a theoretical solution, it represents a viable candidate for real-life systems. Identifying new species in an underwater environment will help us to conquer the unknown and to further advance the human knowledge.

This project aims to help scientists identify unknown marine animals by automatically recognizing possible new marine species in a field setting ie an underwater environment. Therefore, it seeks to address the following research questions: Is it possible to automate unknown marine species' recognition and what an automated system architecture for the recognition of previously unknown marine species could look like. We , thus, proposed a system combining Machine Learning techniques that utilize Convolutional Neural Networks and Novelty Detection. Our system is a form of hybrid intelligence and we also employ supervised learning. Due to the fact that we are aiming to detect unknowns in the form of previously uncatalogued marine animals, in order to validate our approach to this problem we must then apply cross validation. Our system is segregated into two phases: a training phase and a testing or field phase. To approximate the accuracy and legitimacy of our system we must first test it in a laboratory setting.

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A Appendix: Full Interview Transcript with Marcel Walter

Transcript format: Standard Verbatim
Interviewer: Isabella Gardner
Interviewee: Marcel Walter
Date of interview: October 2nd, 2020
Location of interview: Marcel Walter's house
List of Acronyms: IG = Isabella Gardner (interviewer), MW = Marcel Walter (interviewee)

[Begin Transcript 00:00:01]

IG: Hi, my name is Bella, and I am interviewing you on behalf of my Artificial Intelligence project team. I would be grateful to have permission to record this interview.

MW: Yes, gladly. Thanks for having me.

IG: Awesome! Would you mind introducing yourself and telling us a bit about what you've done and currently do?

MW: No, of course not. My name is Marcel Walter. I am a researcher at the University of Bremen in Germany, in the research group of Computer Architecture, and I'm currently a final year Ph.D. student with a specialization in algorithms for the design of emerging technologies.

[00:00:56]

IG: Awesome. So, I am just going to tell you a bit about what our project objectives are and see if you could give us any background information or any advice. The main object of this current project is to discover and classify new marine species by using state-of-the-art technologies such as machine learning algorithms.

There are algorithms that can currently identify different species, such as frogs, butterflies, and birds, through a variety of techniques. There are even underwater species identification that has been done before in shallow waters, searching for a specific class of animals.

However, what we are proposing is an algorithm that can expand the variety of species and aid in the identification of previously undiscovered species, which has never been done. Thus, we were wondering if you had any ideas about what kind of algorithm we could use to solve this problem?

[00:01:48]

MW: That's indeed a very interesting task. So, what you are looking for, if I understood correctly, is you want to record video data, and based on the video data, you want to identify species of animals.

IG: Exactly.

MW: So if you're running an algorithm on video data, then a Machine Learning system like Convolutional Neural Network could be applied, right?

[00:02:18]

IG: I agree. Do you have any other specific suggestions for our system?

[00:04:19]

MW: As you said, the classification of entirely different species has very different attributes to them in their image recognition systems, and therefore discovering or identifying a certain kind of fish from a certain kind of coral might actually be more difficult than just identifying hey this is a fish or it is a coral.

Therefore what you could use is a cascade of different systems like a first kind of pre-processing system which just identifies this is a fish, or this is a coral and then calling the specific Machine Learning system for then dividing it into different kinds of corals, which is then specified for just corals. That could help.

IG: That is really good advice. Do you have any advice on how we could hypothetically maximize its accuracy and not overwhelm it with all these classes?

[00:05:14]

MW: An issue in Machine Learning always is overfitting, which occurs when you're training your Machine Learning algorithm on very specific data so that when it encounters new data, then it cannot handle it anymore. That means it's very good in your test data, but it's not generalizable to the actual environment, which it will be operating in.

The main issue that you're probably facing here is the identification of this unknown class, so you're kind of have to give your system an idea of an unknown class actually is. So it boils down here to the minimization of false positives, where it classifies an animal as some other known animal just because it looks similar.

IG: Okay. So, for instance, what we were thinking about possibly doing is taking our test data and separating it into categories and training our Machine Learning algorithm with test data number one and leaving test number two as unknowns, and so then introducing test data number two after the Machine Learning algorithm has been pre-processed, and so being able to distinguish if it's possible that it can actually identify unknowns.

While this will be previously discovered species, we can then get a better idea if our Machine learning algorithm works. Would you think that it is possible?

[00:06:31]

MW: In the way Machine Learning algorithms work, they have a pre-determined number of classes that they can classify. For instance, if you have, let's say, this made-up number, like fifty different classes of fish that you specifically designed a fish detection algorithm. It will always put a new fish that it sees in any of those fifty classes. What you would like to do then is to have one class that represents unknown fishes, and also for that class, you need training data. But, how would you have this training data if you have never seen an unknown fish? Because that is the definition of unknown.

IG: Exactly.

MW: What else you could indeed do, is to train a network with all these known kinds of fish but, since it will always give an accuracy value, you can say if the accuracy level is lower than the threshold, it's unknown, and thereby you don't have class unknown, but you based it on the accuracy value your Machine Learning system instead.

[00:07:36]

IG: That is actually a very good idea, and we would love to implement it. I would be really grateful if you could look over our flowchart for our proposed algorithm really quickly and see if you have any questions, comments, or improvements.

MW: Sure, I can do that.

[Isabella shows the flowchart to Marcel Walter 00:07:50]

IG: This is it here. We have a testing phase and a training phase. The training phase starts with the input of training images of marine animals that have already been discovered, which we could attain from a database such as OBIS. OBIS is an existing knowledge graph database of all previously discovered marine animals. Then, we would start with the segmentation of the images, feature extraction, classifier training, and in parallel, would be the testing phase, which would go through a pre-processing data step.

For instance, we could use an underwater image processing software, which already exists to try and make it as accurate as possible, improving lighting conditions, improving the reflection of the water. Then, take the pre-processed image and put through the same kind of steps, such as feature extraction and the classification of an image.

We have the classification of things, plants, animals, and unknown animals, though I would not mind incorporating the idea about accuracy. If we get to the end and it still can't tell, then we go back to the retraining step and go back to classifier training. We have a rejection filter at the very end, which outputs a classification as unknown and potentially a new species which would be registered with scientists so they can decide what to do. It is a kind of hybrid intelligence.

Modeled after, for instance, the tumor algorithms which doctors have been using, where tumors are flagged by a Machine learning system, and then sent to doctors to be double-checked. So it's not completely unsupervised; it's mostly supervised learning.

[00:09:52]

MW: I really like the idea of the feedback loop. Feeding newly discovered species into the algorithm to train it on the go, especially because that could prevent you from running into scenarios where you have discovered a new fish indeed, and so your AI system would follow it around for a while taking more pictures for the researchers to analyze them later.

However, the system wouldn't know, wouldn't remember this kind of fish if you wouldn't have the feedback loop.

Therefore, if it encounters the same species again or potentially the same individual again, it follows it around and thereby wasting time and resources, which you could better use by saying, oh yeah, I've seen that fish before because I've retrained my model, and therefore I flag this fish as known now. I really like that step of your algorithm.

[00:10:43]

IG: Awesome. Thank you. This is all just a theoretical project but, would you mind telling us in your opinion how possible you think this could be and when something similar could be implemented? Just a rough estimation obviously. I mean, I know this is partially outside of your expertise because it has a lot to do also with alpha taxonomy, marine biology, etc. I would love just your prediction or opinion.

MW: As you said, I am not a marine biologist, and I do not know how hard it actually is to classify certain kinds of fish or other marine species. So I don't know how much they are distinguished by their features that you can extract from images taking underwater. Being unsure about this part, I can, as you said, only give a rough estimation.

Machine Learning algorithms are getting better and better, especially with the advancement in Convolutional Neural Networks. We have image classification algorithms which are really good and can these days be even incorporated into streams of image data as we see in autonomous cars for instance, that have to analyze terabytes of data every day, which is pretty cool. So, I don't see why an underwater vehicle with lights and sensors should not be able to detect and classify some marine animals.

I do not know if it will reach accuracies of ninety percent or higher, which is typically what you want. You want as close to one hundred percent already, of course. I do not know if it will be achievable by the system. That depends a lot on the training data that you have but since you told me that this OBIS database, which apparently contains a lot of data about marine animals and species already, I think this might be a feasible project. I don't see why it shouldn't be funded.

[00:12:54]

IG: Awesome. Thank you very much, and thank you for time.

MW: You're welcome.

[End Transcript 00:12:58]

B Appendix: Full Interview Transcript with Juan Gardner

Transcript format: Standard Verbatim
Interviewer: Isabella Gardner
Interviewee: Juan Gardner
Date of interview: October 10th, 2020
Location of interview:Isabella Gardner's house
List of Acronyms: IG = Isabella Gardner (interviewer), JG = Juan Gardner (interviewee)

IG: Hi, my name is Bella and I am interviewing you on behalf of my artificial intelligence project team. I would be grateful to have permission to record this interview. Would you mind introducing yourself and telling us a bit about what you have done and currently do?

JG: Good Day, my name is Juan Gardner and I am currently the Technical Manager for Fugro Marine Germany GmbH, a multinational company providing services for geo-intelligence and asset integrity solutions for large constructions, infrastructure and natural resources. My company here in Germany is specifically associated with doing hydrographic work for marine charting as well full geophysical and geotechnical survey for subsea cables (fiber optic (internet) and power (offshore wind energy)). I have been in the offshore survey business for a little over 30 years.

IG: The main objective of this project is to discover and classify new marine species by using state of the art technologies like machine learning algorithms. There are algorithms that can identify different species such as frogs, butterflies, and birds through a variety of techniques4. Even underwater species identification has been done before in shallow waters, searching for a specific class of animals. However, what we are proposing is an algorithm that can span a variety of species and aid in the identification of previously undiscovered species! What kind of vehicles autonomous or not could host our AI?

JG: As these observations can be conducted from both the surface as well as submerged, there are quite a few different vehicles available. For current fishery management the most stereotypical is the larger science / research vessels holding the larger sonars... in my mind these platforms would be your best starting point for proof of concept... given the types of sonar in use... this is very scalable... larger vessel use typically lower frequency /lower resolution sonars but these have the largest foundation of learning materials... where as the smaller AUVs use higher frequency sonars which are more precise... but having less available learning material.

IG: What kind of AI is currently being utilized in marine exploration?

JG:AI is very exciting technology / approach for my field of endeavors... however it is very much in the 'beginning days' The most typical use so far is for editing; removing sonar noise from hydrographic data sets, and for target classification; finding specific targets e.g. boulders meeting a certain criteria from thousands 6 thousands of cataloged targets. The use of AI has also been adapted to improving our navigational data… similar 'smart' noise filtering.

IG: How original is our project?

JG:The use of AI for fisheries is new to me but I would believe already a topic somewhere… mainly this is an onerous task that is well suited to automation where much is subjective classification now. From my limited experience, I would guess the main issue is developing enough learning material to well train you AI.

IG:Where do you think AI could be used in this ocean?

JG:In theory almost anywhere, there are many low carbon footprint platforms already collecting various data autonomously… however more typical is where you have larger vessel deploy multiple small AUVs as a force multiplier.

IG: How autonomous and widespread are AUVs?

JG:For my point of view (within the industry) these are basically available globally. However, the very large and more sophisticated units probably number less that a thousand globally… and many of these assigned to the various military. However nearly every single one, both large and small, have capacities or already carries versions of sonar that could be used in this application.